



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Lecture 5

Molecular Geometry and Bonding Theories (Chapter9)

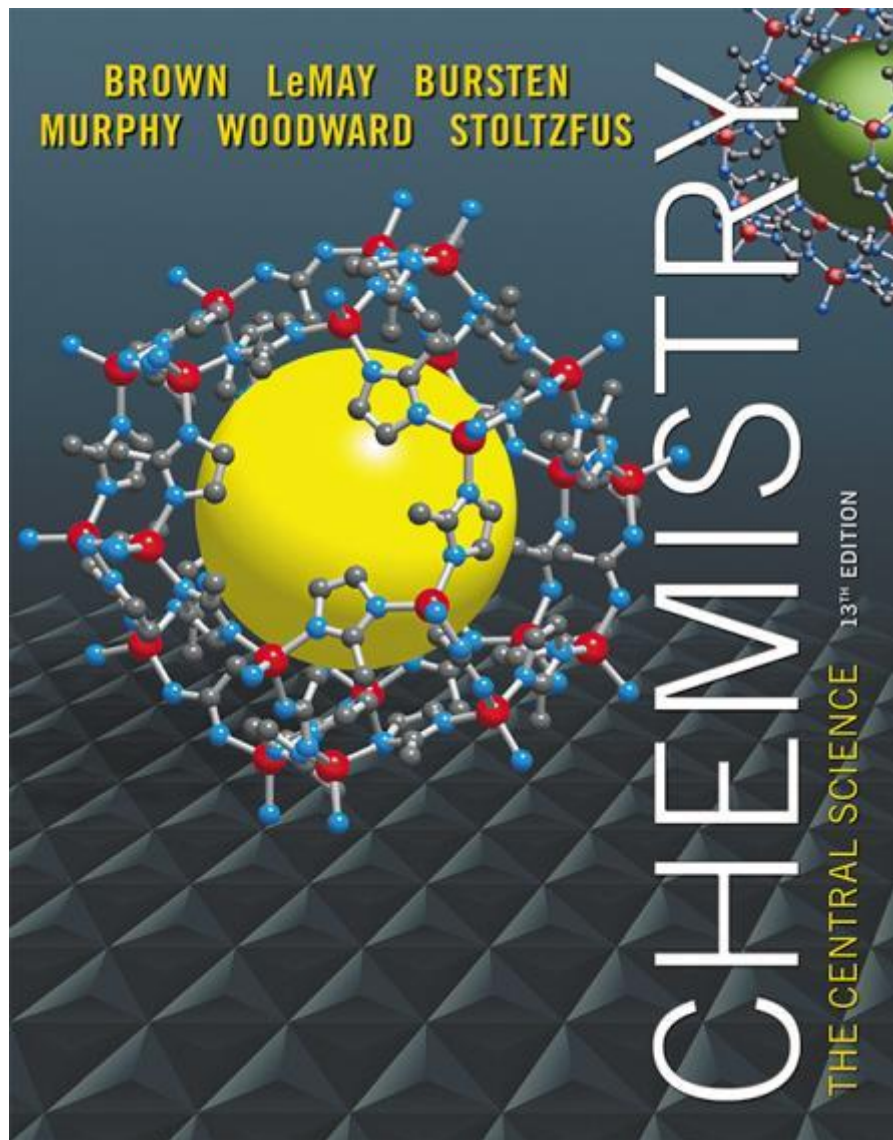
Instructor: Prof. Daigo Miyajima

E-mail: dmiyajima@cuhk.edu.cn

Office: RX1101

TA: Yang Jialin, Yida Wang, Lu Yixin,
Hongchen Wan, Nanting Qiu,
Jiaxian Wang,

Sep 16th, 2025



Lecture Presentation

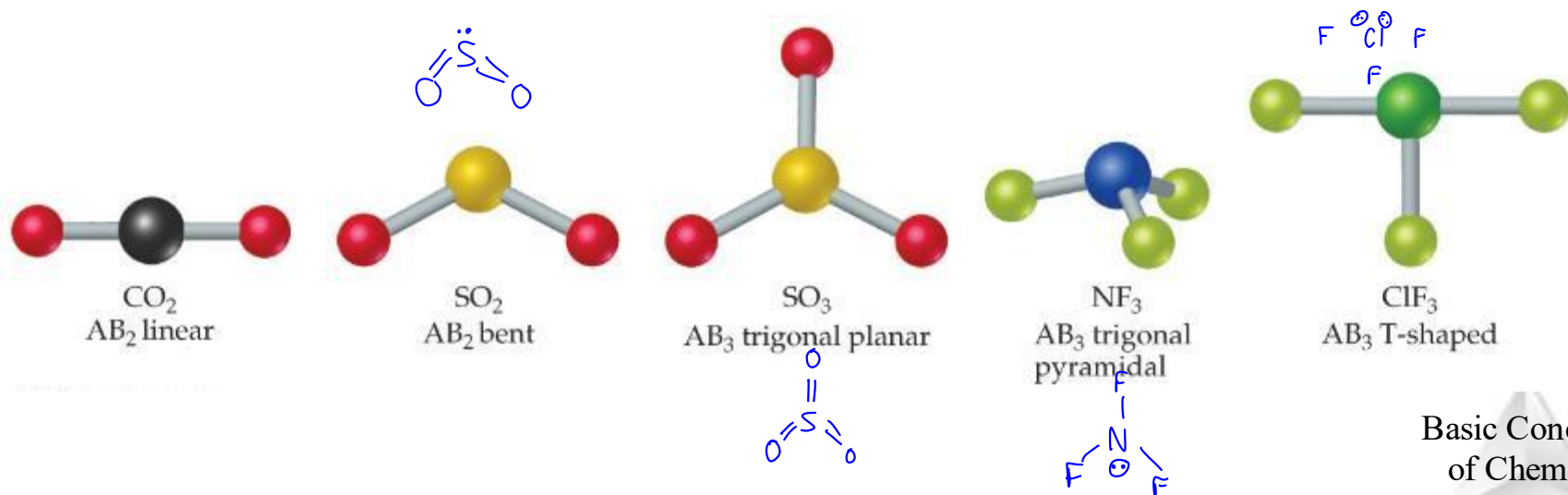
Chapter 9

Molecular Geometry and Bonding Theories

Basic Concepts
of Chemical
Bonding

Molecular Shapes

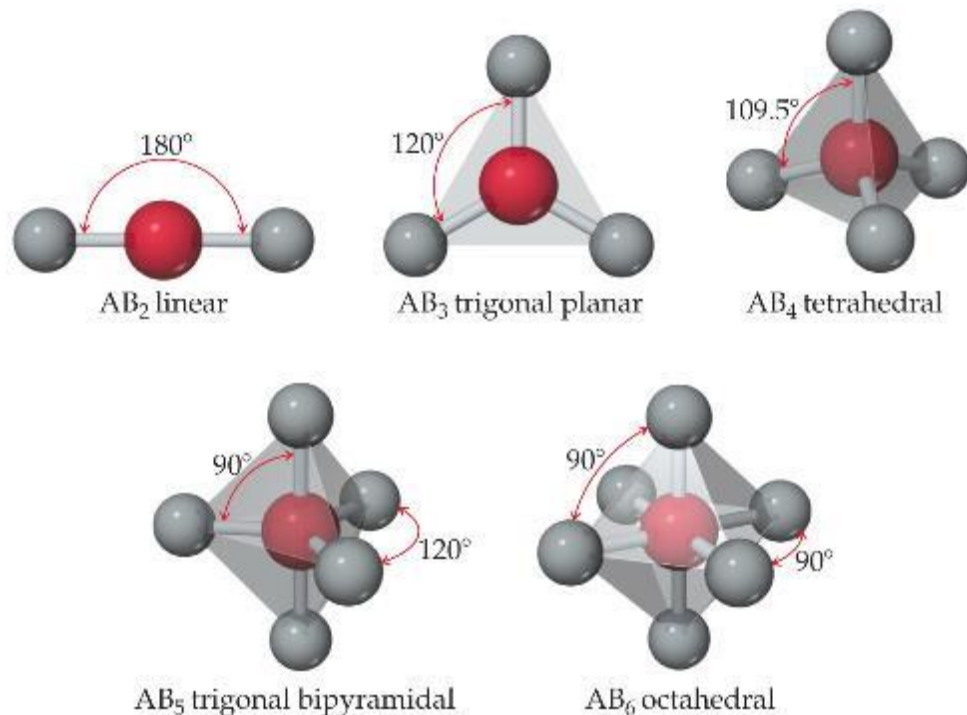
- Lewis Structures show bonding and lone pairs, but do *not* denote shape.
- However, we *use* Lewis Structures to help us determine shapes.
- Here we see some common shapes for molecules with two or three atoms connected to a central atom.



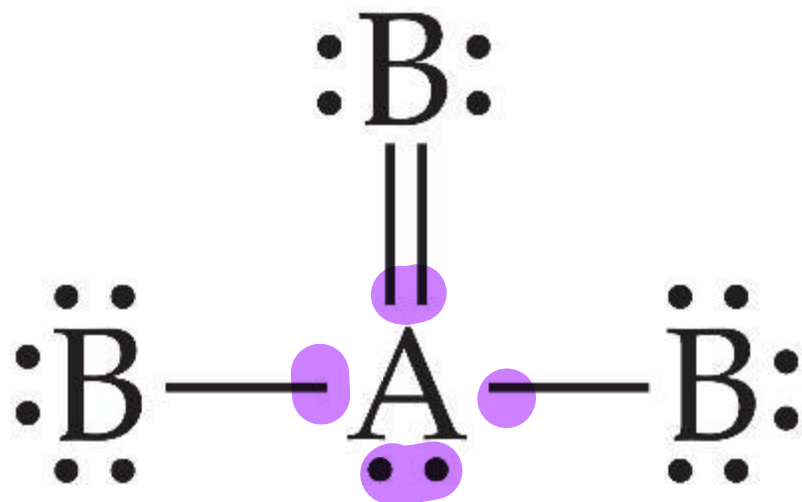
Basic Concepts
of Chemical
Bonding

What Determines the Shape of a Molecule?

- **Electron pairs**, whether bonding or nonbonding, **repel each other**.
- Assuming they arrange themselves as far apart as possible allows us to predict the molecule's shape
- This is the Valence-Shell Electron-Pair Repulsion (**VSEPR**) model.

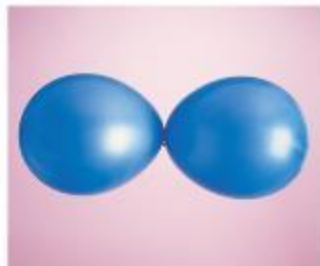


Electron Domains



- The **directions in which electron pairs** are located are called **electron domains**. This applies whether the domain contains **a single pair of electrons or multiple pairs in the same direction**.
- In this molecule, the central atom **A** has **four electron domains**, as determined by its **bonding pairs and lone pairs**.

Valence-Shell Electron-Pair Repulsion (VSEPR) Model



Two balloons
linear orientation



Three balloons
trigonal-planar orientation



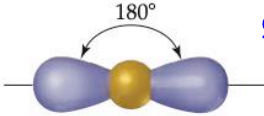
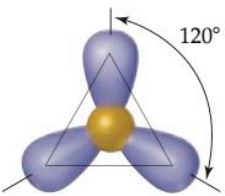
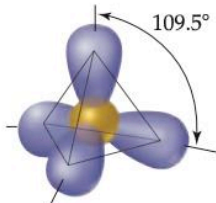
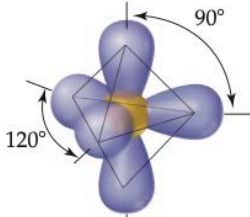
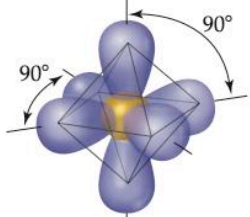
Four balloons
tetrahedral orientation

*“The best arrangement of a given number of electron domains is the one that **minimizes the repulsions** among them.”*

(The balloon analogy in the figure to the left demonstrates the maximum distances, which minimize repulsions.)

Electron-Domain Geometries

Table 9.1 Electron-Domain Geometries as a Function of Number of Electron Domains

Number of Electron Domains	Arrangement of Electron Domains	Electron-Domain Geometry	Predicted Bond Angles
2		Linear	180°
3		Trigonal planar	120°
4		Tetrahedral	109.5°
5		Trigonal bipyramidal	120° 90°
6		Octahedral	90°

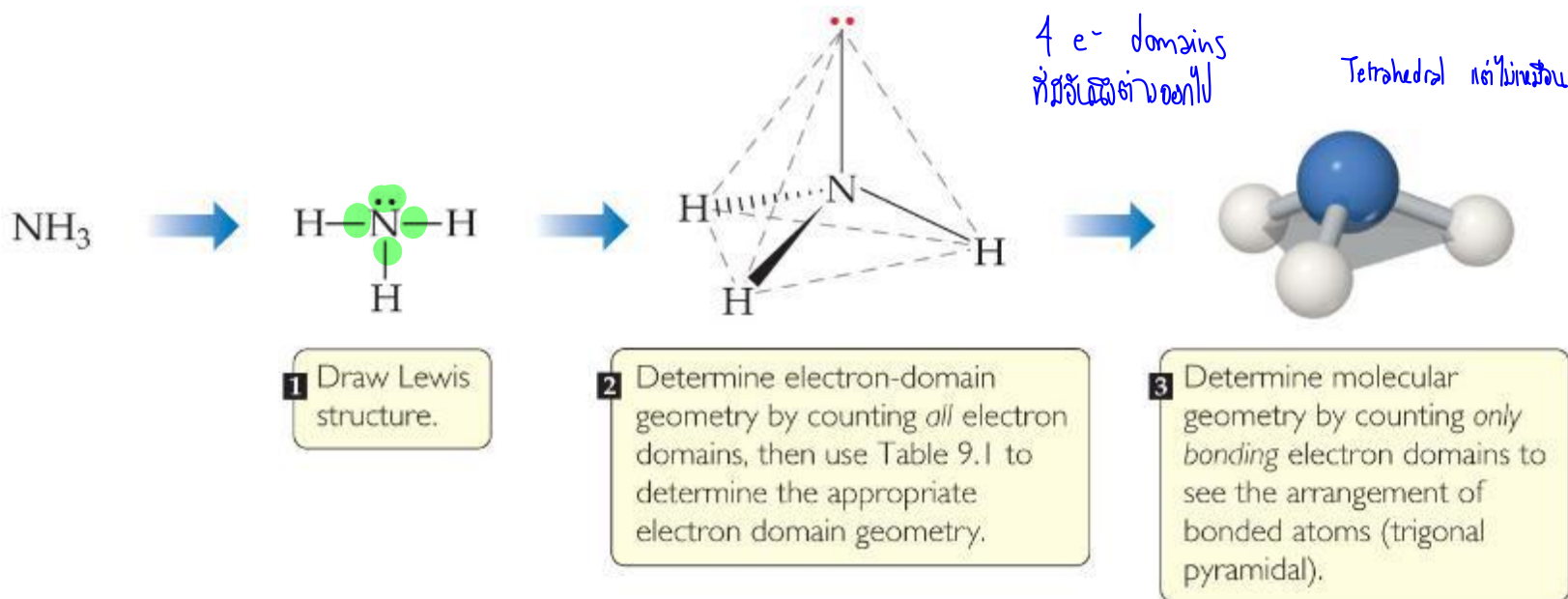
2 e⁻ domains minimize repulsion → linear

รูปทรงสามมิติ

- The Table shows the electron-domain geometries for two through six electron domains around a central atom.

- To determine the electron-domain geometry, count the total number of lone pairs, single, double, and triple bonds on the central atom.

Molecular Geometries



- Once you have determined the electron-domain geometry, use the arrangement of the bonded atoms to determine the **molecular geometry**.
- Tables 9.2 and 9.3 show the potential molecular geometries. We will look at each electron domain to see what molecular geometries are possible.

Linear Electron Domain

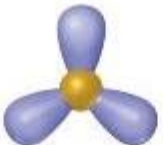
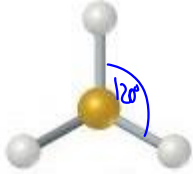
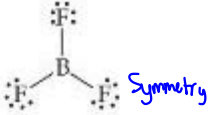
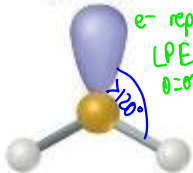
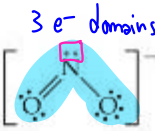
Table 9.2 Electron-Domain and Molecular Geometries for Two, Three, and Four Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
2	 Linear	2	0	 Linear	$\text{:}\ddot{\text{O}}=\text{C}=\ddot{\text{O}}\text{:}$

- In the linear domain, there is only one molecular geometry: linear.
- NOTE: If there are only two atoms in the molecule, the molecule will be linear no matter what the electron domain is.

Trigonal Planar Electron Domain

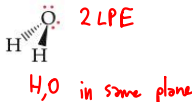
Table 9.2 Electron-Domain and Molecular Geometries for Two, Three, and Four Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
3	 Trigonal planar	3	0 AX_3E_0	 Trigonal planar	 Symmetry
		2	1 AX_2E_1	 Bent	 3 e ⁻ domains e ⁻ repulsion now LPE increases bond angle < 120° Same plane

- There are two molecular geometries:
 - trigonal planar, if all electron domains are bonding, and
 - bent, if one of the domains is a nonbonding pair.

Tetrahedral Electron Domain

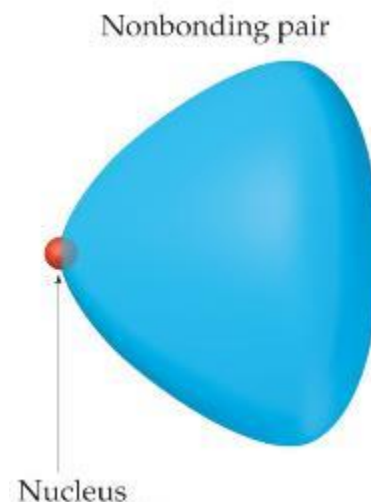
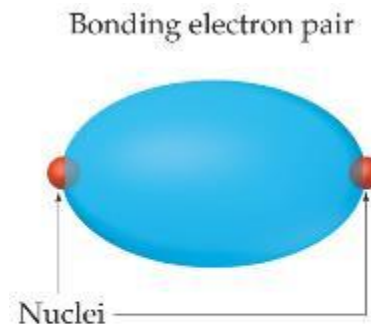
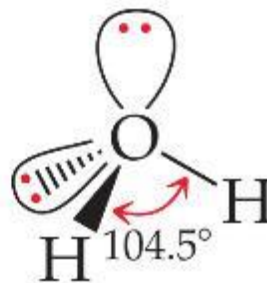
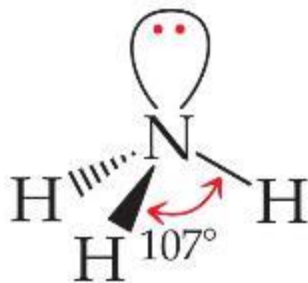
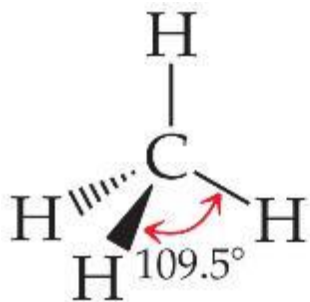
Table 9.2 Electron-Domain and Molecular Geometries for Two, Three, and Four Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
4	 Tetrahedral	4	0	 Tetrahedral	
		3	1	 Trigonal pyramidal	
		2	2	 Bent	

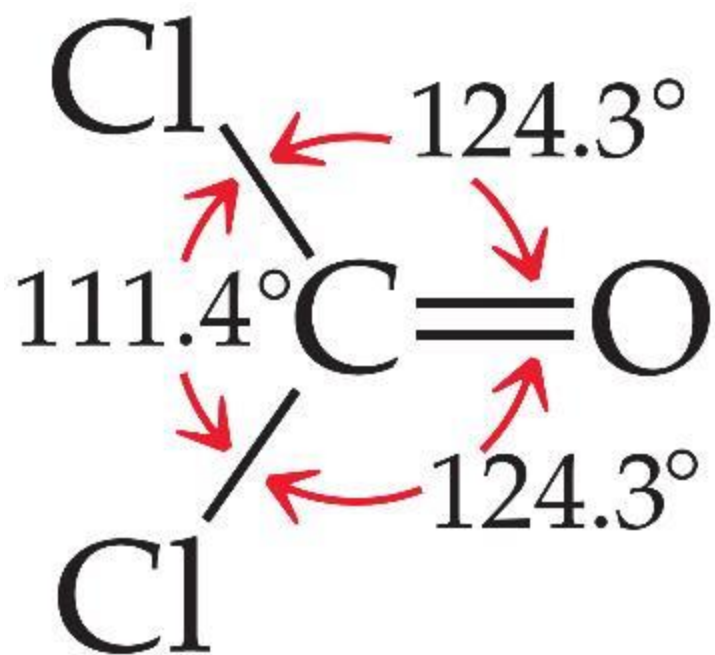
- There are three molecular geometries:
 - tetrahedral, if all are bonding pairs,
 - trigonal pyramidal, if one is a nonbonding pair, and
 - bent, if there are two nonbonding pairs.

Nonbonding Pairs and Bond Angle

- Nonbonding pairs are physically larger than bonding pairs.
- Therefore, their repulsions are greater; this tends to compress bond angles.



Multiple Bonds and Bond Angles

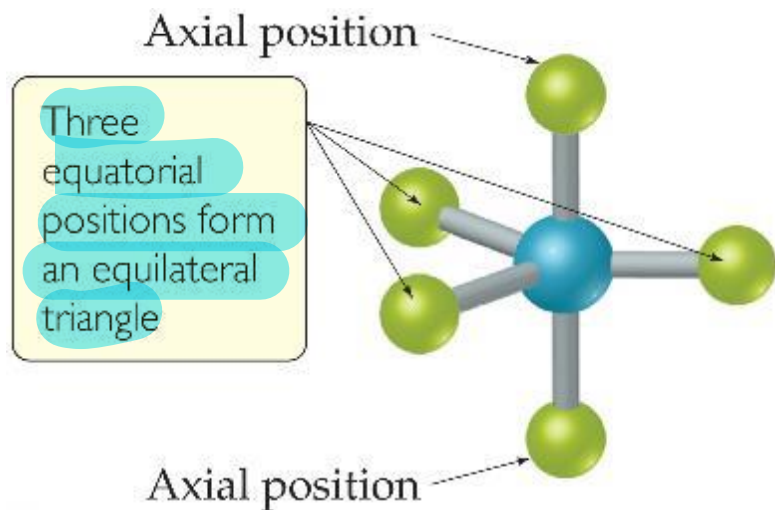


- Double and triple bonds have larger electron domains than single bonds.
- They exert a greater repulsive force than single bonds, making their bond angles greater.

Expanding beyond the Octet Rule

- Remember that some elements can break the octet rule and make *more* than four bonds (or have more than four electron domains).
- The result is two more possible electron domains: five = trigonal bipyramidal; six = octahedral (as was seen in the slide on electron-domain geometries).

Trigonal Bipyramidal Electron Domain



- There are two distinct positions in this geometry:

- Axial
- Equatorial

- Lone pairs occupy equatorial positions

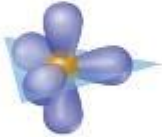
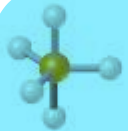
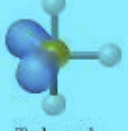
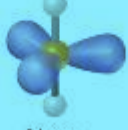
(Repulsion between two domains is much greater when they are situated 90° from each other than when they are at 120°). *Minimize repulsion*

Basic Concepts
of Chemical
Bonding

Trigonal Bipyramidal Electron Domain

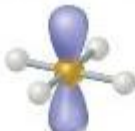
- There are four distinct molecular geometries in this domain:
 - Trigonal bipyramidal
 - Seesaw
 - T-shaped
 - Linear

Table 9.3 Electron-Domain and Molecular Geometries for Five and Six Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
5	 Trigonal bipyramidal	5	0	 Trigonal bipyramidal	PCl_5
		4	1	 Seesaw	SF_4
		3	2	 T-shaped	ClF_3
		2	3	 Linear	XeF_2

Octahedral Electron Domain

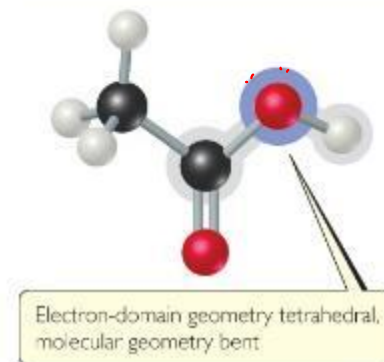
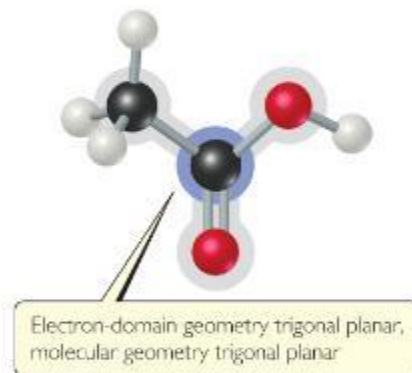
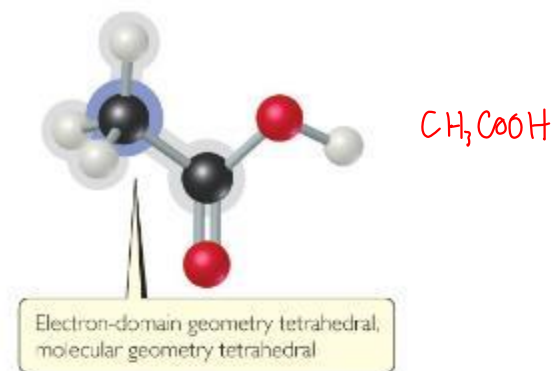
Table 9.3 Electron-Domain and Molecular Geometries for Five and Six Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
6	 Octahedral	6	0	 Octahedral	SF_6
		5	1	 Square pyramidal	BrF_5
		4	2	 Square planar	XeF_4

- All positions are equivalent in the octahedral domain.
- There are three molecular geometries:
 - Octahedral
 - Square pyramidal
 - Square planar

Shapes of Larger Molecules

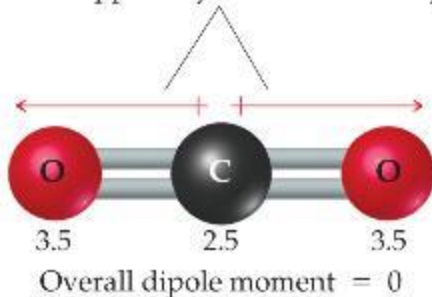
For larger molecules,
look at the geometry
about each atom
rather than the
molecule as a whole.



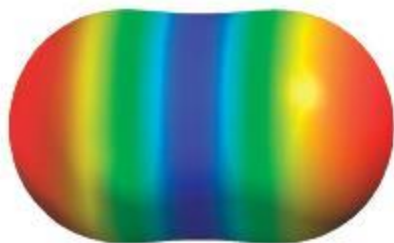
Comparison of the Polarity of Two Molecules

A NONPOLAR molecule

Equal and oppositely directed bond dipoles



ต้องดูว่า dipole
หักล้างกันหมดไหม



Low electron density

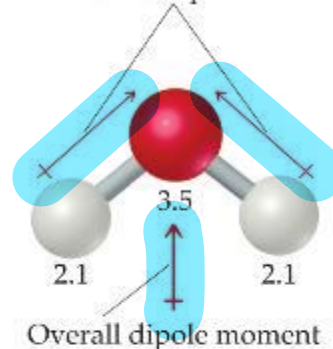
High electron density

+

-

A POLAR molecule

Bond dipoles



Overall dipole moment



Low electron density

High electron density

+

-

Basic Concepts
of Chemical
Bonding

Polarity of Molecules

Ask yourself:

COVALENT or IONIC? If COVALENT:

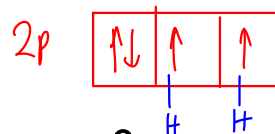
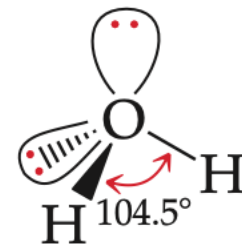
Are the BONDS polar?

- a. NO: The molecule is NONPOLAR!
- b. YES: Continue—Do the AVERAGE position of $\delta+$ and $\delta-$ coincide?
 - 1) YES: The molecule is NONPOLAR.
 - 2) NO: The molecule is POLAR.

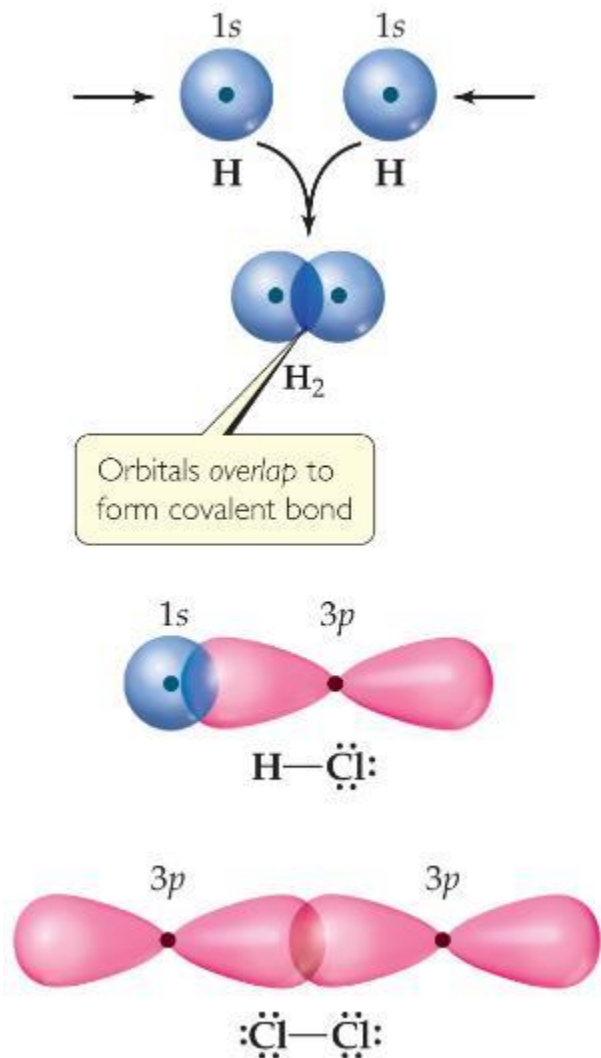
NOTE: When different atoms are bonded to the central atom, the varying bond polarities often break molecular symmetry, making the molecule likely polar.

VSEPR and Hybrid Orbitals

- VSEPR predicts shapes of molecules very well.
- How does that fit with orbitals?
- Let's use H_2O as an example:
- If we draw the best Lewis structure to assign VSEPR, it becomes bent.
- If we look at oxygen, its electron configuration is $1s^2 2s^2 2p^4$. If it shares two electrons to fill its valence shell, they should be in $2p$.
- Wouldn't that make the angle 90° ?
- Why is it 104.5° ?



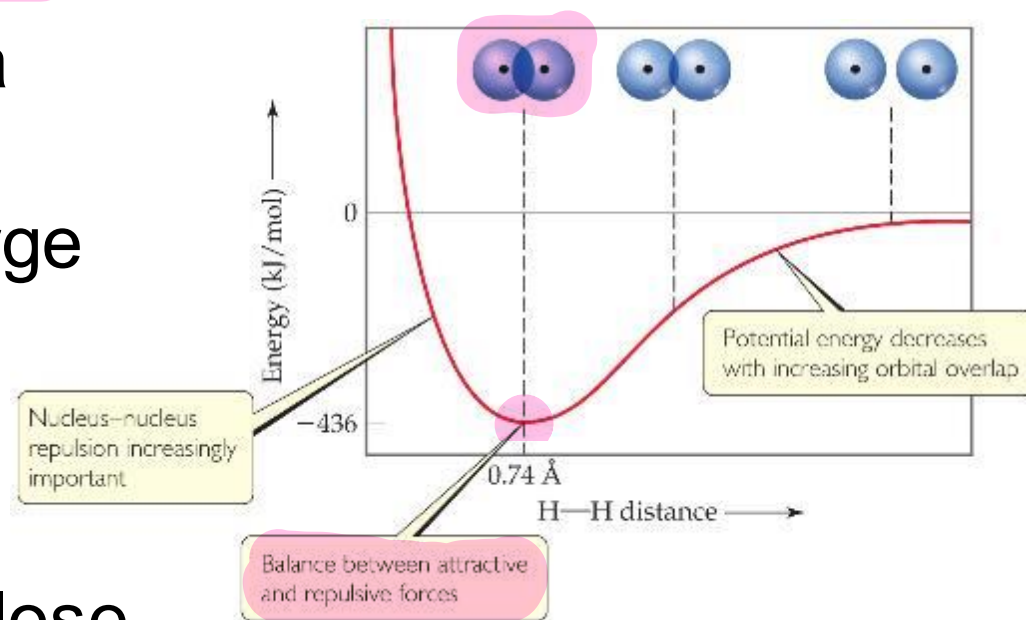
Valence-Bond Theory



- In **Valence-Bond Theory**, **electrons of two atoms begin to occupy the same space.**
- This is called “overlap” of orbitals.
- The sharing of space between **two electrons of opposite spin** results in a **covalent bond.**

Overlap and Bonding

- Increased overlap brings the electrons and nuclei closer together until a balance is reached between the like charge repulsions and the electron-nucleus attraction.
- Atoms can't get too close because the internuclear repulsions get too great.

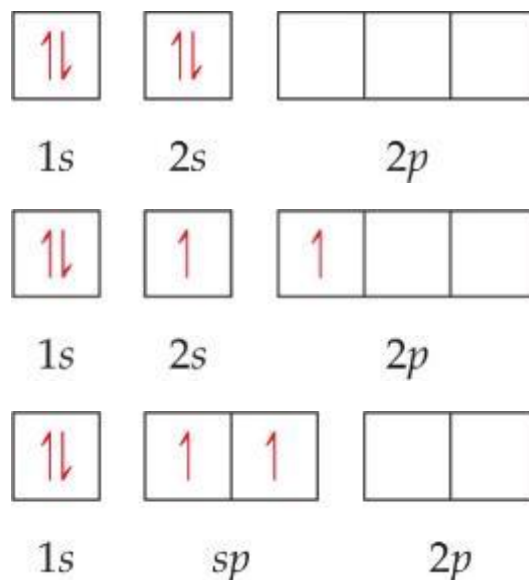


Hybrid Orbitals

- **Hybrid orbitals** form by “mixing” of atomic orbitals to create new orbitals of equal energy, called degenerate orbitals.
- When two orbitals “mix” they create two orbitals; when three orbitals mix, they create three orbitals; etc.

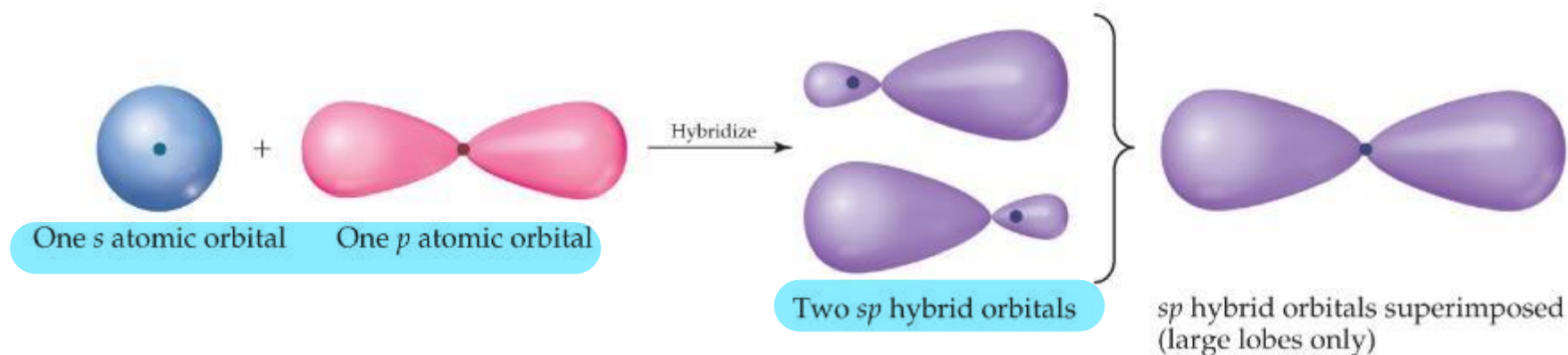
Be—*sp* hybridization

- When we look at the orbital diagram for beryllium (Be), we see that there are only paired electrons in full sub-levels.
- Be makes electron deficient compounds with two bonds for Be. Why? *sp* hybridization (mixing of one *s* orbital and one *p* orbital)



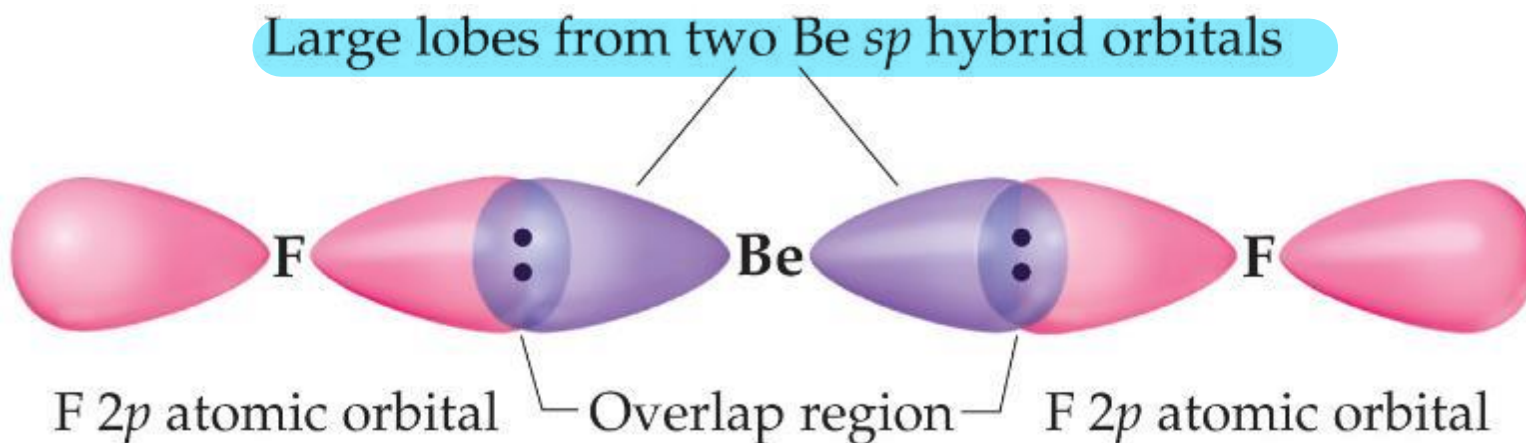
sp Orbitals

- Mixing the s and p orbitals yields two degenerate orbitals that are hybrids of the two orbitals.
 - These sp hybrid orbitals have two lobes like a p orbital.
 - One of the lobes is larger and more rounded, as is the s orbital.



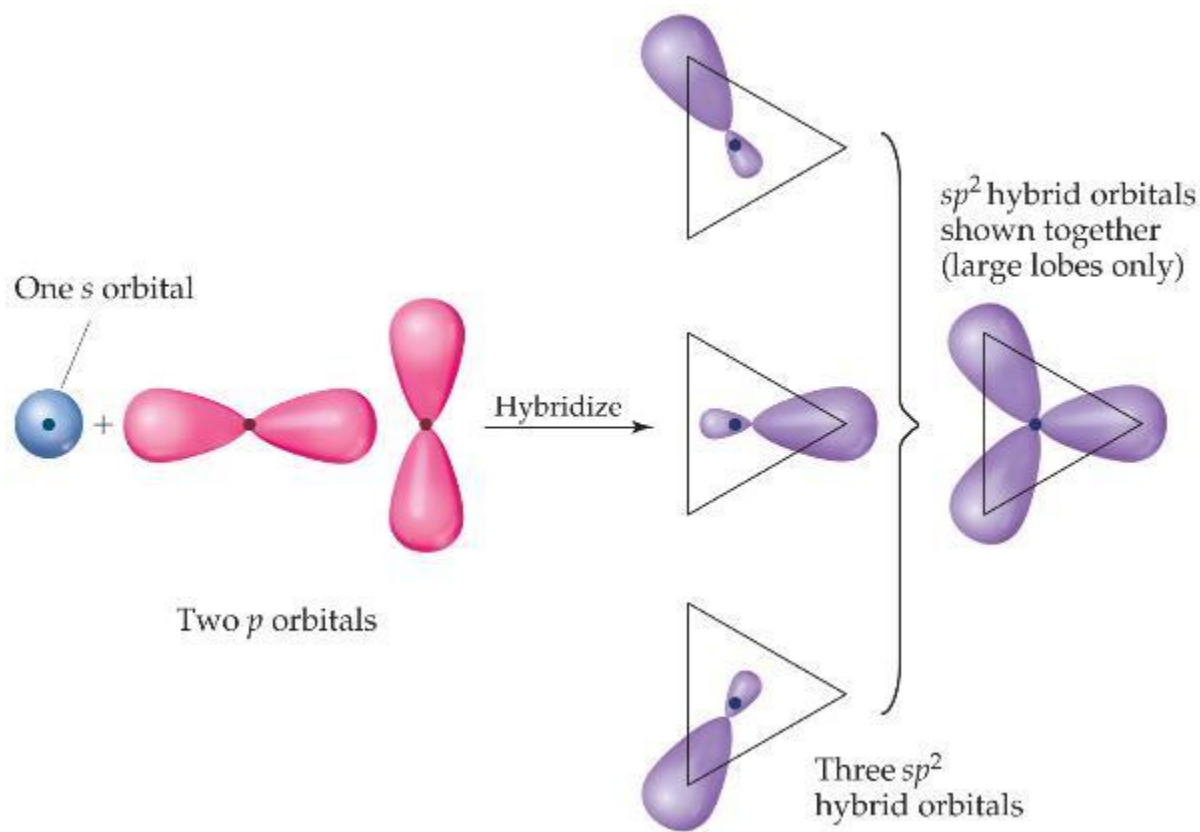
Position of sp Orbitals

- These two degenerate orbitals would align themselves 180° from each other.
- This is consistent with the observed geometry of Be compounds (like BeF_2) and VSEPR: linear.



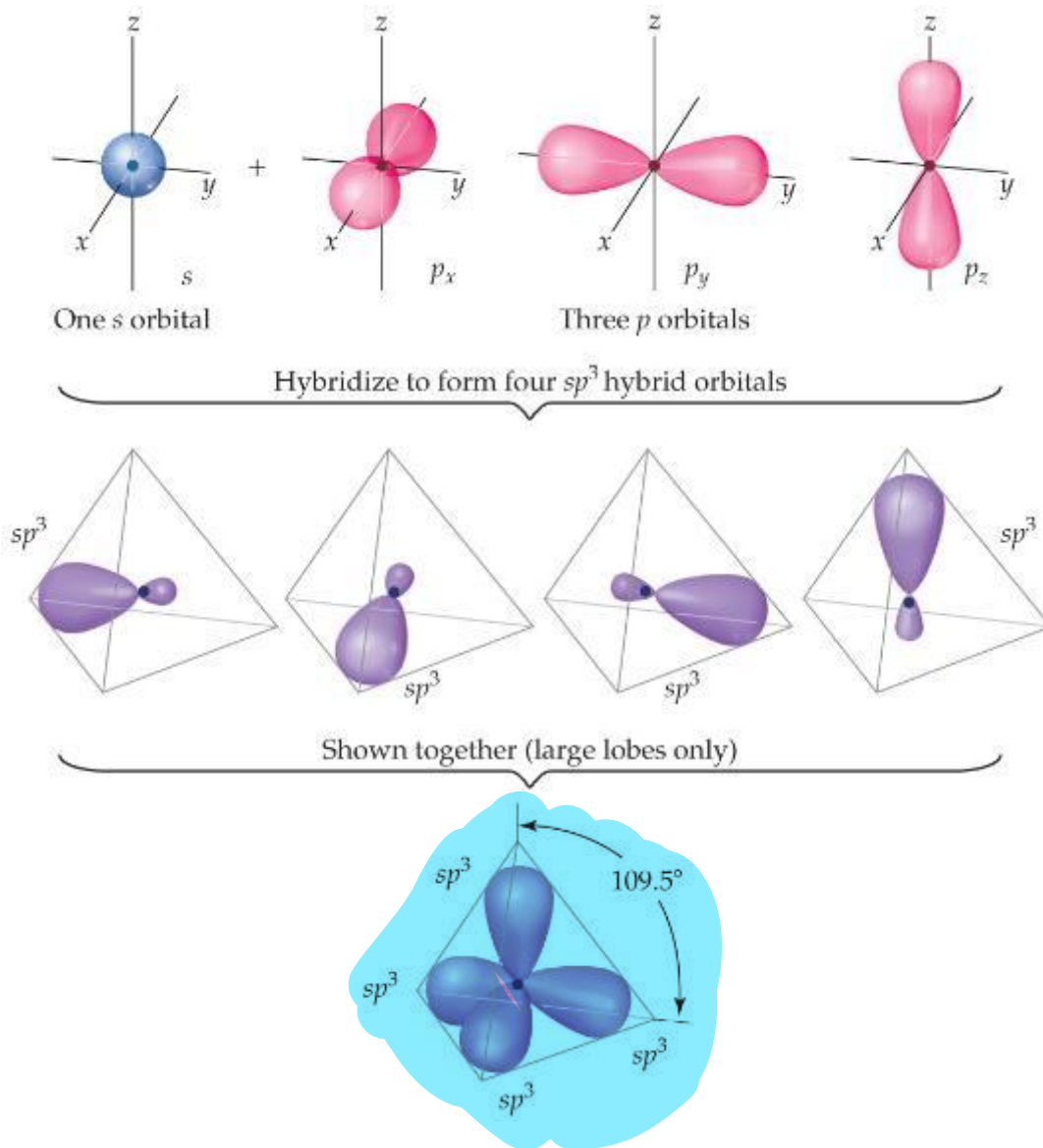
Boron—Three Electron Domains Gives sp^2 Hybridization

Using a similar model for boron leads to three degenerate sp^2 orbitals.



Carbon: sp^3 Hybridization

With carbon,
we get four
degenerate
 sp^3 orbitals.

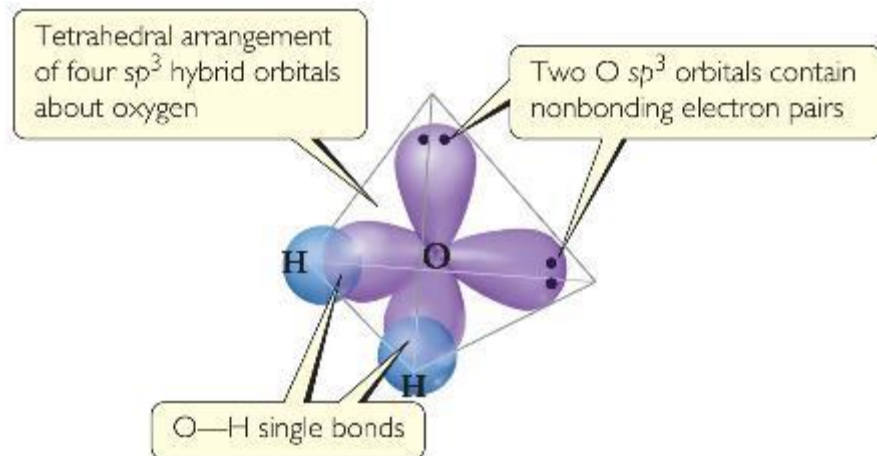


Hypervalent Molecules

- The elements which have *more than an octet*
- Valence-Bond model would use *d* orbitals to make more than four bonds.
- *This view works for period 3 and below.*
- Theoretical studies suggest that the energy needed would be too great for this.
- A more detailed bonding view is needed than we will use in this course.

What Happens with Water?

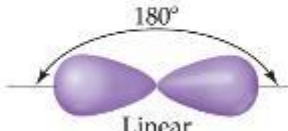
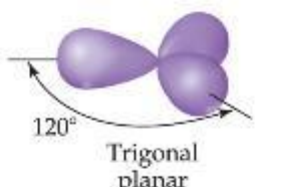
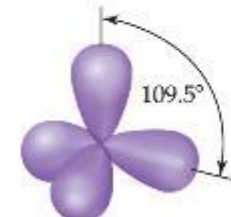
- We started this discussion with H_2O and the angle question: Why is it 104.5° instead of 90° ?
- Oxygen has two bonds and two lone pairs—four electron domains.
- The result is sp^3 hybridization!



Hybrid Orbital Summary

- 1) Draw the Lewis structure.
- 2) Use VSEPR to determine the electron-domain geometry.
- 3) Specify the hybrid orbitals needed to accommodate these electron pairs.

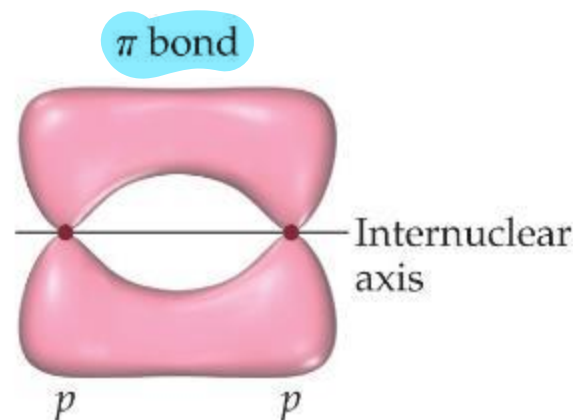
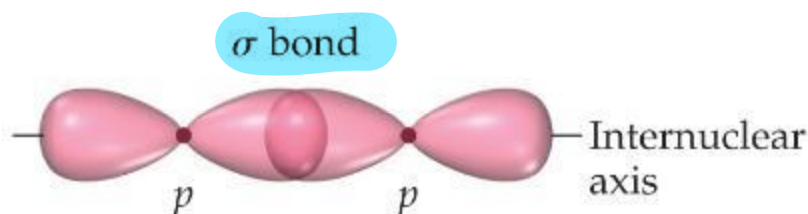
Table 9.4 Geometric Arrangements Characteristic of Hybrid Orbital Sets

Atomic Orbital Set	Hybrid Orbital Set	Geometry	Examples
s, p	Two sp	 Linear	$\text{BeF}_2, \text{HgCl}_2$
s, p, p	Three sp^2	 Trigonal planar	BF_3, SO_3
s, p, p, p	Four sp^3	 Tetrahedral	$\text{CH}_4, \text{NH}_3, \text{H}_2\text{O}, \text{NH}_4^+$

Types of Bonds

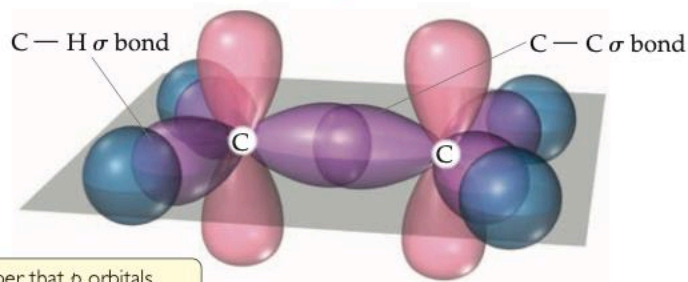
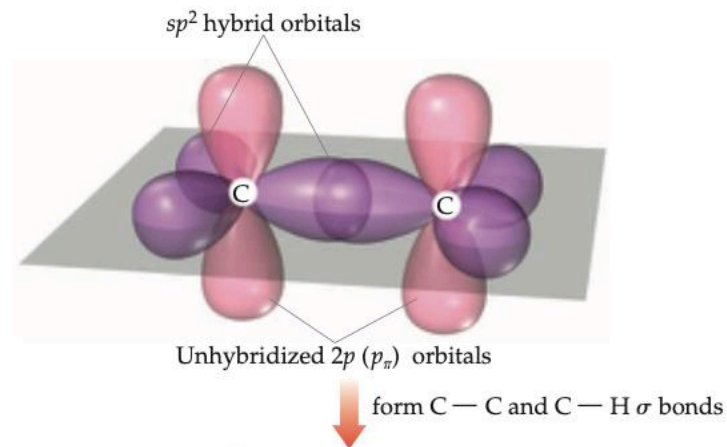
- How does a double or triple bond form?
- It *can't*, if we only use hybridized orbitals.
- *However*, if we use the orbitals which are *not* hybridized, we can have a “side-ways” overlap.
- Two types of bonds:
- Sigma (σ) bond
- Pi (π) bond

Sigma (σ) and Pi (π) Bonds

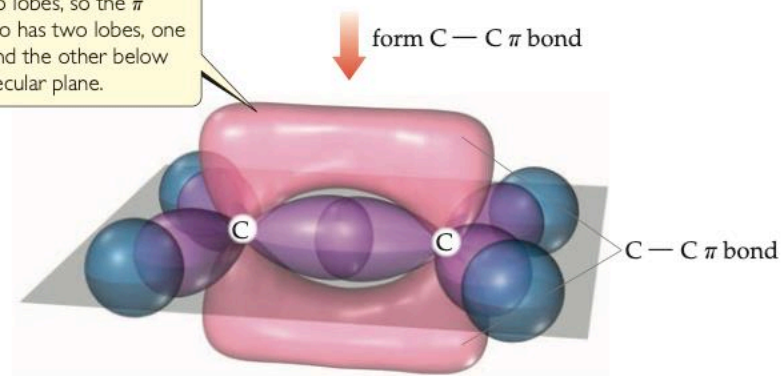


- Sigma bonds are characterized by
 - head-to-head overlap.
 - cylindrical symmetry of electron density about the internuclear axis.
- Pi bonds are characterized by
 - side-to-side overlap.
 - electron density above and below the internuclear axis.

Bonding in Molecules



Remember that p orbitals have two lobes, so the π bond also has two lobes, one above and the other below the molecular plane.

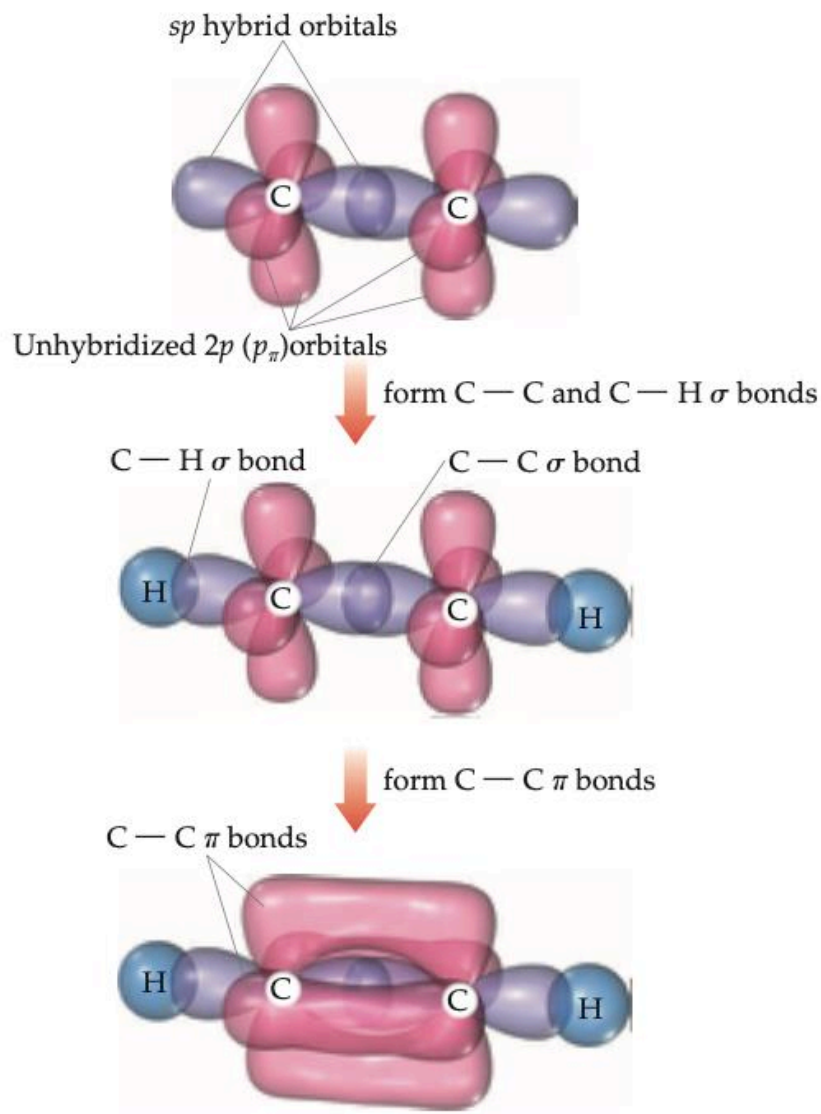


- Single bonds are always σ -bonds.
- Multiple bonds have one σ -bond, all other bonds are π -bonds.



Bonding in Molecules

- Single bonds are always σ -bonds.
- Multiple bonds have one σ -bond, all other bonds are π -bonds.

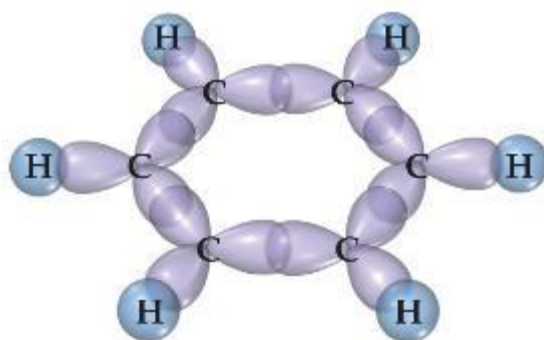


Localized or Delocalized Electrons

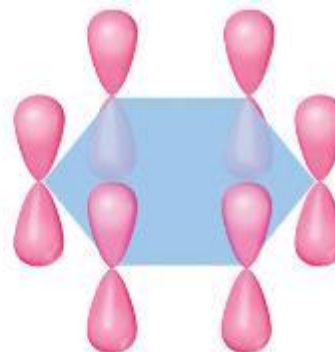
- Bonding electrons (σ or π) that are specifically shared between two atoms are called **localized electrons**.
- In many molecules, we can't describe all electrons that way (resonance); the other electrons (shared by multiple atoms) are called **delocalized electrons**.

Benzene

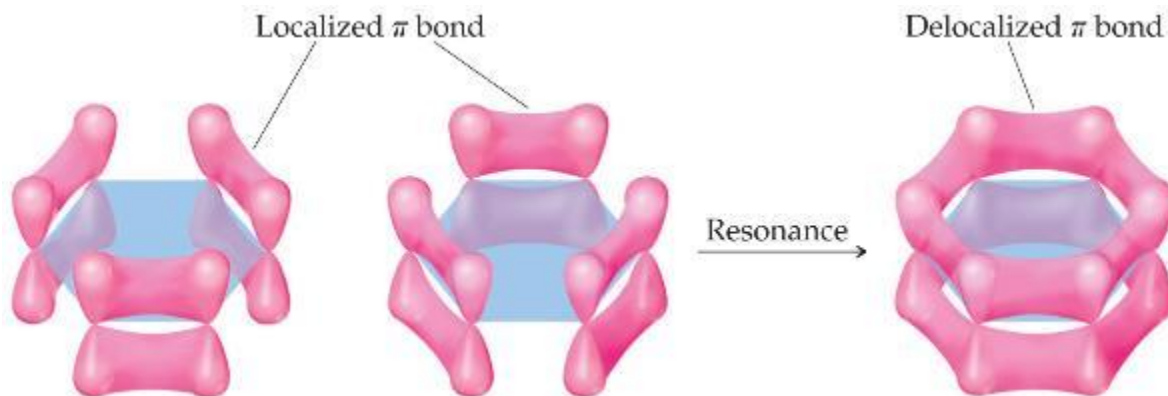
The organic molecule benzene (C_6H_6) has six σ -bonds and a p orbital on each C atom, which form delocalized bonds using one electron from each p orbital.



(a) σ bonds



(b) p_π orbitals



Molecular Orbital (MO) Theory

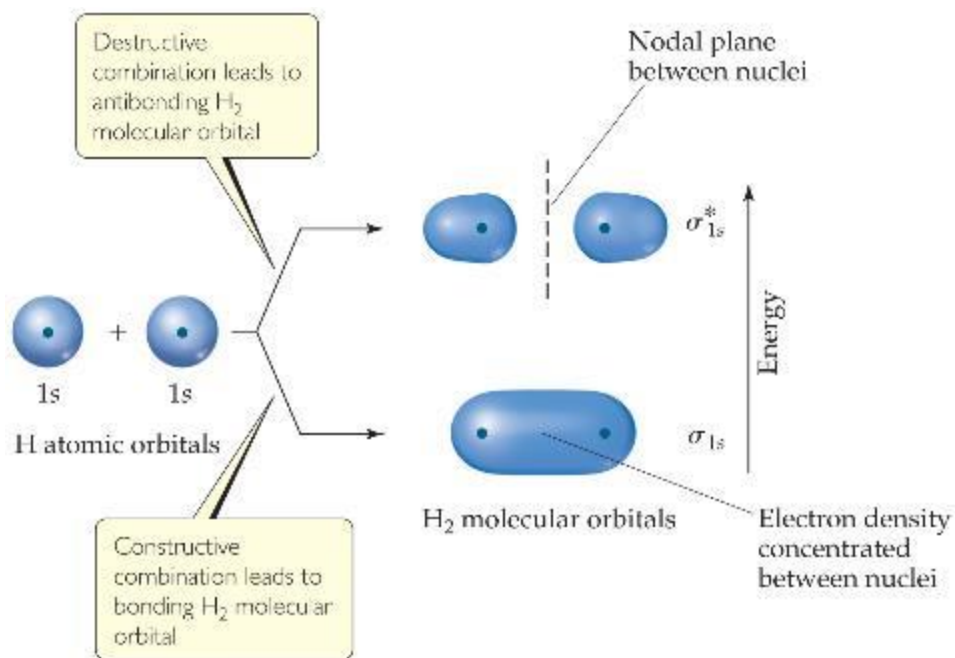
- Wave properties are used to describe the energy of the electrons in a molecule.
- **Molecular orbitals** have many characteristics like atomic orbitals:
 - maximum of two electrons per orbital
 - Electrons in the same orbital have opposite spin.
 - Definite energy of orbital
 - Can visualize electron density by a contour diagram

More on MO Theory

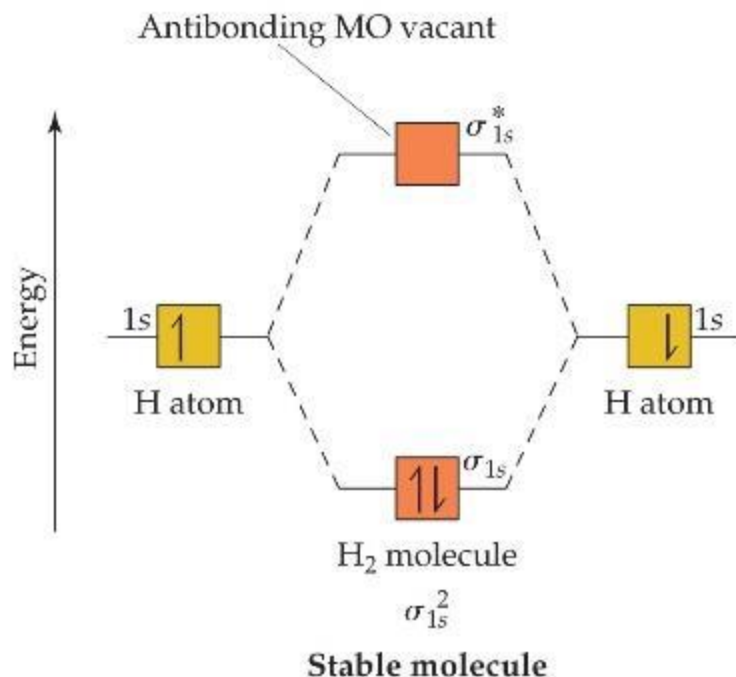
- They differ from atomic orbitals because they represent the entire molecule, not a single atom.
- Whenever two atomic orbitals overlap, two molecular orbitals are formed: one bonding, one antibonding.
- **Bonding orbitals** are constructive combinations of atomic orbitals.
- **Antibonding orbitals** are destructive combinations of atomic orbitals. They have a new feature unseen before: A **nodal plane** occurs where electron density equals zero.

Molecular Orbital (MO) Theory

Whenever there is direct overlap of orbitals, forming a bonding and an antibonding orbital, they are called **sigma (σ) molecular orbitals**. The antibonding orbital is distinguished with an asterisk as σ^* . Here is an example for the formation of a hydrogen molecule from two atoms.



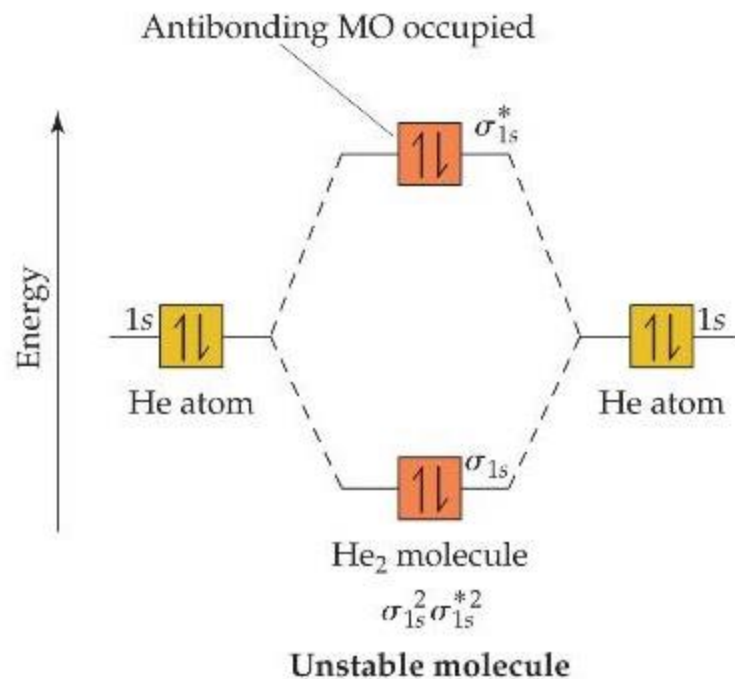
MO Diagram



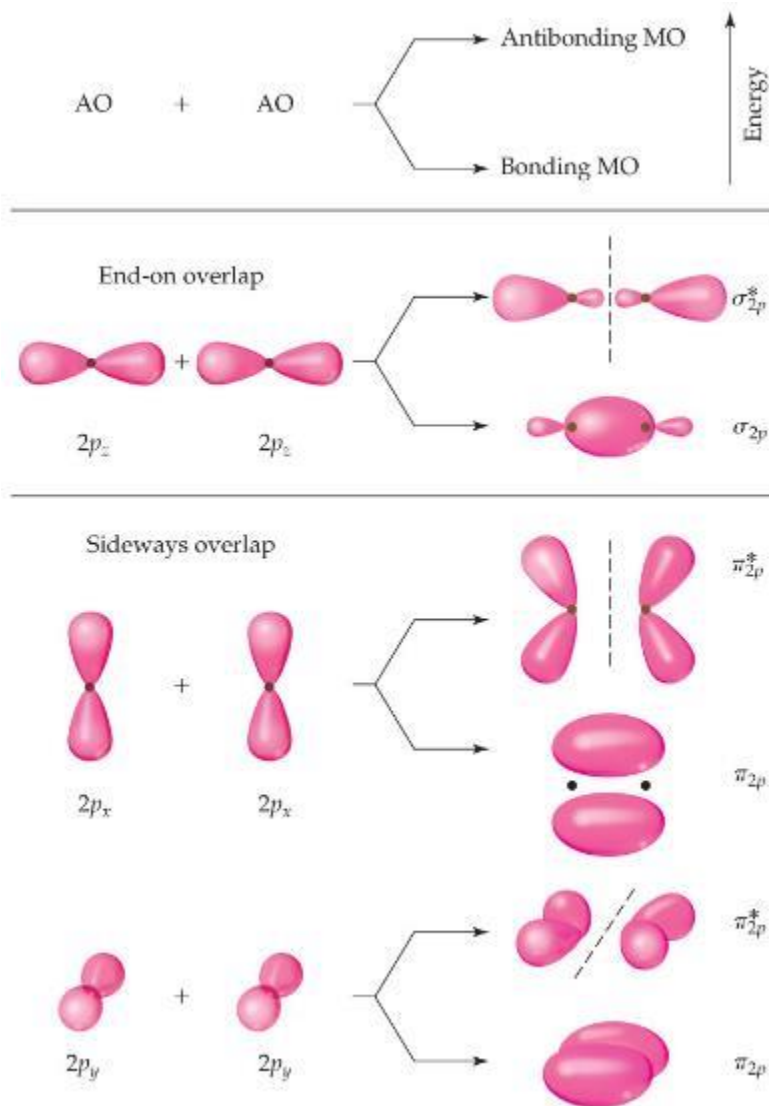
- An **energy-level diagram**, or **MO diagram** shows how orbitals from atoms combine to give the molecule.
- In H₂ the two electrons go into the bonding molecular orbital (lower in energy).
- **Bond order** = $\frac{1}{2}(\text{\# of bonding electrons} - \text{\# of antibonding electrons}) = \frac{1}{2}(2 - 0) = 1 \text{ bond}$

Can He₂ Form? Use MO Diagram and Bond Order to Decide!

- Bond Order = $\frac{1}{2}(2 - 2)$
= 0 bonds
- Therefore, He₂ does *not* exist.



s and p Orbitals Can Interact

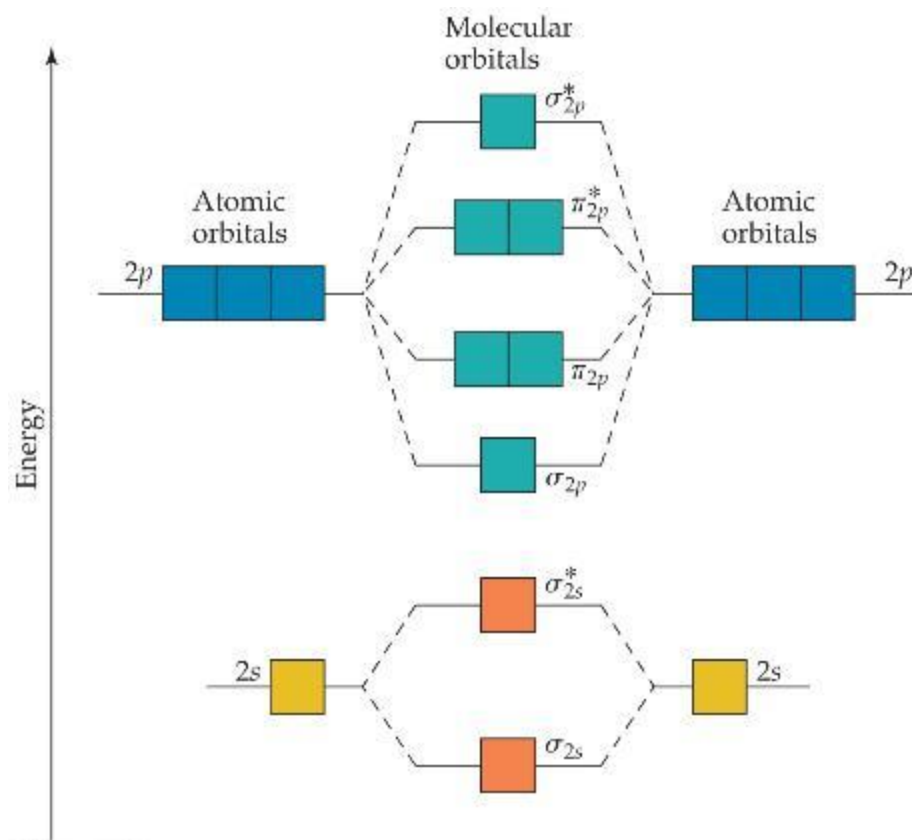


- For atoms with both s and p orbitals, there are two types of interactions:
- The s and the p orbitals that face each other overlap in σ fashion.
- The other two sets of p orbitals overlap in π fashion.
- These are, again, direct and “side-ways” overlap of orbitals.

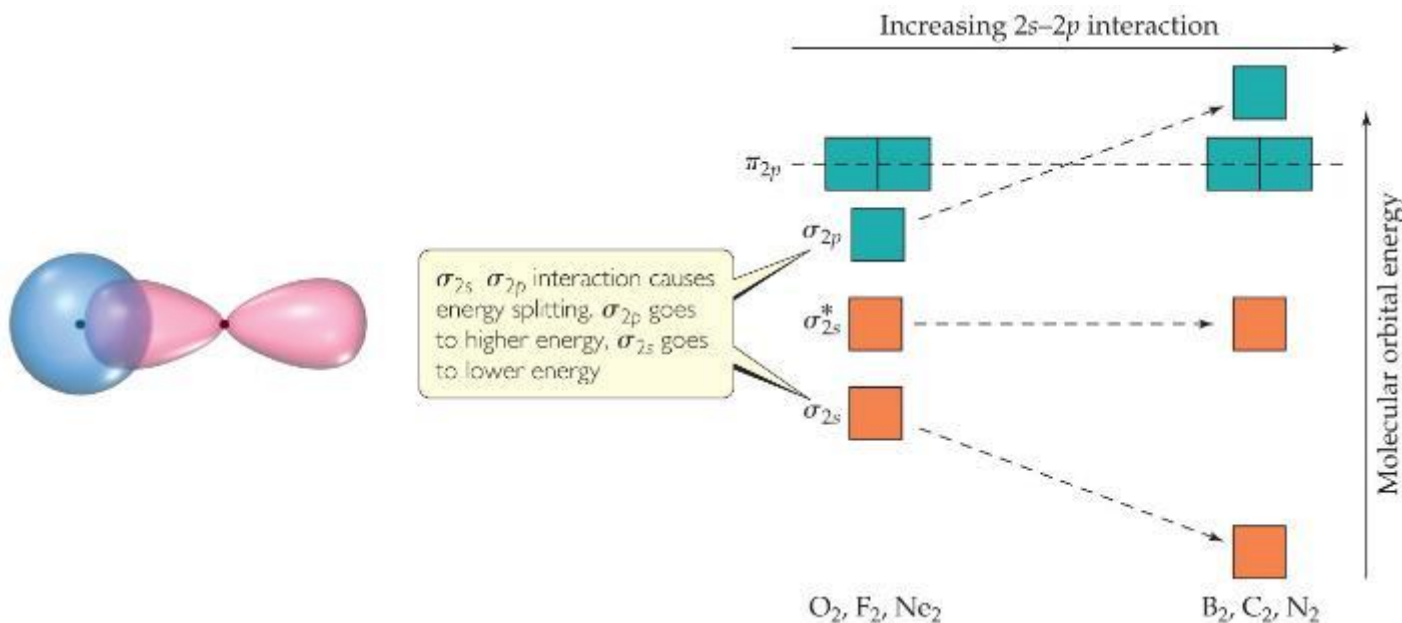
MO Theory

- The resulting MO diagram:

- There are σ and σ^* orbitals from s and p atomic orbitals.
- There are π and π^* orbitals from p atomic orbitals.
- Since direct overlap is stronger, the effect of raising and lowering energy is greater for σ and σ^* .

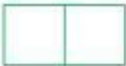
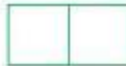
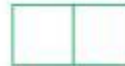
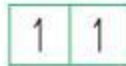
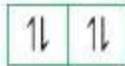
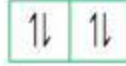
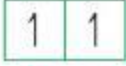
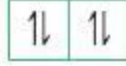
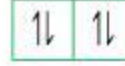


s and p Orbital Interactions



- In some cases, s orbitals can interact with the p_z orbitals more than the p_x and p_y orbitals.
- It raises the energy of the p_z orbital and lowers the energy of the s orbital.
- The p_x and p_y orbitals are degenerate orbitals.

MO Diagrams for Diatomic Molecules of 2nd Period Elements

	Large 2s–2p interaction			Small 2s–2p interaction		
	B ₂	C ₂	N ₂	O ₂	F ₂	Ne ₂
σ_{2p}^*						
π_{2p}^*						
σ_{2p}						
π_{2p}						
σ_{2s}^*						
σ_{2s}						
Bond order	1	2	3	2	1	0
Bond enthalpy (kJ/mol)	290	620	941	495	155	—
Bond length (Å)	1.59	1.31	1.10	1.21	1.43	—
Magnetic behavior	Paramagnetic	Diamagnetic	Diamagnetic	Paramagnetic	Diamagnetic	—

MO Diagrams and Magnetism

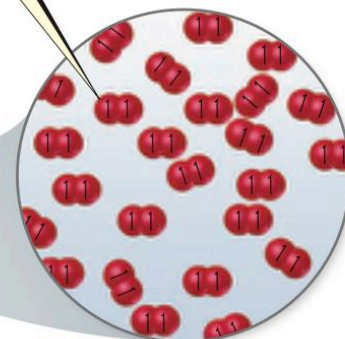
- **Diamagnetism** is the result of all electrons in every orbital being spin paired. These substances are weakly repelled by a magnetic field.
- **Paramagnetism** is the result of the presence of one or more unpaired electrons in an orbital.
- Is oxygen (O_2) paramagnetic or diamagnetic? Look back at the MO diagram! It is paramagnetic.

Paramagnetism of Oxygen

- Lewis structures would *not* predict that O_2 is paramagnetic.
- The MO diagram clearly shows that O_2 is paramagnetic.
- Both show a double bond (bond order = 2).



Because O_2 molecules are paramagnetic ...



... they are attracted into the magnetic field.

Heteronuclear Diatomic Molecules

- Diatomic molecules can consist of atoms from different elements.
- How does a MO diagram reflect differences?
- The atomic orbitals have different energy, so the interactions change slightly.
- The more electronegative atom has orbitals lower in energy, so the bonding orbitals will more resemble them in energy.

